MARSEL BAKASHOV

Master's Student in Machine Learning
Data Science / ML / RL / AI

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ABOUT ME

Machine learning master's student with a software engineering background and internships in healthcare, insurance and media. Experienced in reinforcement learning, AI agents and large-scale semantic search.

SKILLS

Programming: Python, SQL, C++, Java, Bash, R
ML / Data: PyTorch, Scikit-learn, TensorFlow, HuggingFace, MLflow, RL, AI Agents
Tools: Azure, Databricks, Snowflake, Power BI, Airflow, Docker, Git
Languages: English, French, Russian, Spanish, Korean

EDUCATION

1/2026 – 1/2028 **Professional Master's in Machine Learning** — ***University of Montreal / McGill - MILA***
GPA: 4.0 / 4.0 (in progress) — Montreal, Canada

9/2021 – 12/2025 **Bachelor of Software Engineering** — ***Polytechnique Montreal***
GPA: 3.4 / 4.0 — Montreal, Canada

2/2025 – 6/2025 **Bachelor of Computer Engineering** — ***Korea University College of Engineering***
Academic Exchange — Seoul, South Korea

EXPERIENCE

5/2026 – 12/2026 **Data Science & ML Intern** — ***McKesson***
- Developed AI agents used by 100+ team members to analyze business data and automate analytical workflows, saving hours of manual work.
- Processed sensitive health data (PHI) via API integrations, prioritizing privacy and security.
- Produced executive summaries from analytical findings to support decision-making.

`Python / AI Agents / APIs / Azure / Snowflake / Power BI / Claude Code`

9/2025 – 4/2026 **Data Scientist & RL Intern** — ***Intact Insurance***
- Developed reinforcement learning pricing policies that improved simulated long-term profitability under regulatory and business constraints, using adaptive primal–dual optimization.
- Validated policies in simulation on profitability, volatility and CVaR risk to guide pricing decisions.

`Python / PyTorch / Reinforcement Learning / CVaR / MLflow / Databricks`

5/2024 – 9/2024 **Data Scientist Intern** — ***Radio-Canada / CBC***
- Designed an AI-powered indexing system unifying internal content sources for semantic and keyword search across 15M+ elements, used by 100+ employees.

`Python / Azure AI Search / ELK / Airflow / Embeddings / Hugging Face`

9/2023 – 12/2023 **Full-Stack Developer Intern** — ***Stingray***
- Developed full-stack features in an Agile team, with analytics, automated testing, CI and code reviews.

`Next.js / React / Node.js / TypeScript`

PROJECTS

Personal Project **GravityGuyML — Reinforcement Learning Side-Scroller** — ***GitHub***
- Built a Gymnasium-compatible Pygame environment with procedural generation and deep RL agents (`DQN`, `PPO`, `SAC`, `A2C`).

`Python / Deep RL / Gymnasium / TensorFlow / Pygame`

Personal Project **Quantitative Finance Notes — Monte Carlo & GBM Modeling** — ***GitHub***
- Built reproducible notebooks on stochastic processes, Monte Carlo simulations, and Geometric Brownian Motion for market modeling and data calibration.

`Python / NumPy / Pandas / Monte Carlo / GBM`

Academic Project **Weather-Based Disease Outbreak Prediction** — ***GitHub***
- Predicted outbreaks from climate data using classification models (Logistic Regression, SVM, Decision Trees) with F1-score and AUC-ROC evaluation.

`Python / scikit-learn / pandas / FP-Growth`

EXTRACURRICULAR ACTIVITIES

1/2024 – 8/2024 **Teaching Assistant – Discrete Mathematics** — ***Polytechnique Montreal***
- Led review sessions, graded assignments, and supported 80+ students.

1/2024 – 5/2024 **Sponsorship Director** — ***PolyAI Hackathon***
- Raised over $22,000 in funding and led a sponsorship team with weekly meetings.